

800V Three-Level Flying Capacitor Traction Inverter

Design and Comprehensive Loss Analysis
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Project Overview

Design Choice: 800V Architecture

To meet modern EV performance demands, an **800V DC bus architecture** was explicitly chosen for this design over traditional 400V systems to address three primary vehicle-level objectives:

Halved Phase Currents: Doubling the bus voltage from 400V to 800V halves the required phase current for the exact same motor power output.

Reduced Ohmic Losses: Lower currents drastically reduce I^2R conduction losses in the traction inverter switches, motor stator windings, and high-voltage DC cabling.

Ultra-Fast Charging: An 800V bus allows for DC fast charging at power levels exceeding 350 kW without requiring prohibitively thick, heavy, or liquid-cooled charging cables.

The dv/dt Problem and the Multilevel Solution

The 800V Challenge

A standard 2-level VSI at 800V switches the full bus voltage in under 50 ns, generating **dv/dt spikes exceeding 16 kV/μs**.

Standard motor insulation is rated for ~1200 V/μs maximum — the **limit is exceeded by over 13×**.

This causes partial discharge and **insulation breakdown within 1000 hours** of continuous operation.

The Multilevel Solution

A 3-level topology reduces the voltage step to **V_{dc}/2 = 400V**, halving dv/dt to ~8 kV/μs.

The improved waveform **reduces output THD** by synthesizing a closer sine approximation.

Lower harmonic content reduces EMI and allows a **smaller, lighter output filter**.

Topology Selection: FC vs NPC

Neutral Point Clamped (NPC) Limitations

Unequal loss distribution between inner and outer switches leads to asymmetric heating and complex cooling requirements.

Neutral point balance is notoriously difficult to maintain under **transient EV loads**, requiring complex control algorithms.

Flying Capacitor (FC) Advantages

All 4 switches block identical voltage ($V_{dc}/2 = 400V$), ensuring **symmetric thermal stress** and allowing identical power modules.

Redundant switching states enable **natural capacitor balancing** regardless of the load power factor.

Trade-off: Requires a bulky capacitor and pre-charge circuit, which is acceptable in modern EVs where pre-charge contactors already exist.

FC Switching States and PD-PWM

3-Phase Structure: Each phase leg (a, b, c) utilizes four switches (e.g., q_{1a} , q_{2a} , $\bar{q}_{2a}$, $\bar{q}_{1a}$) with a flying capacitor bridging the inner junctions.

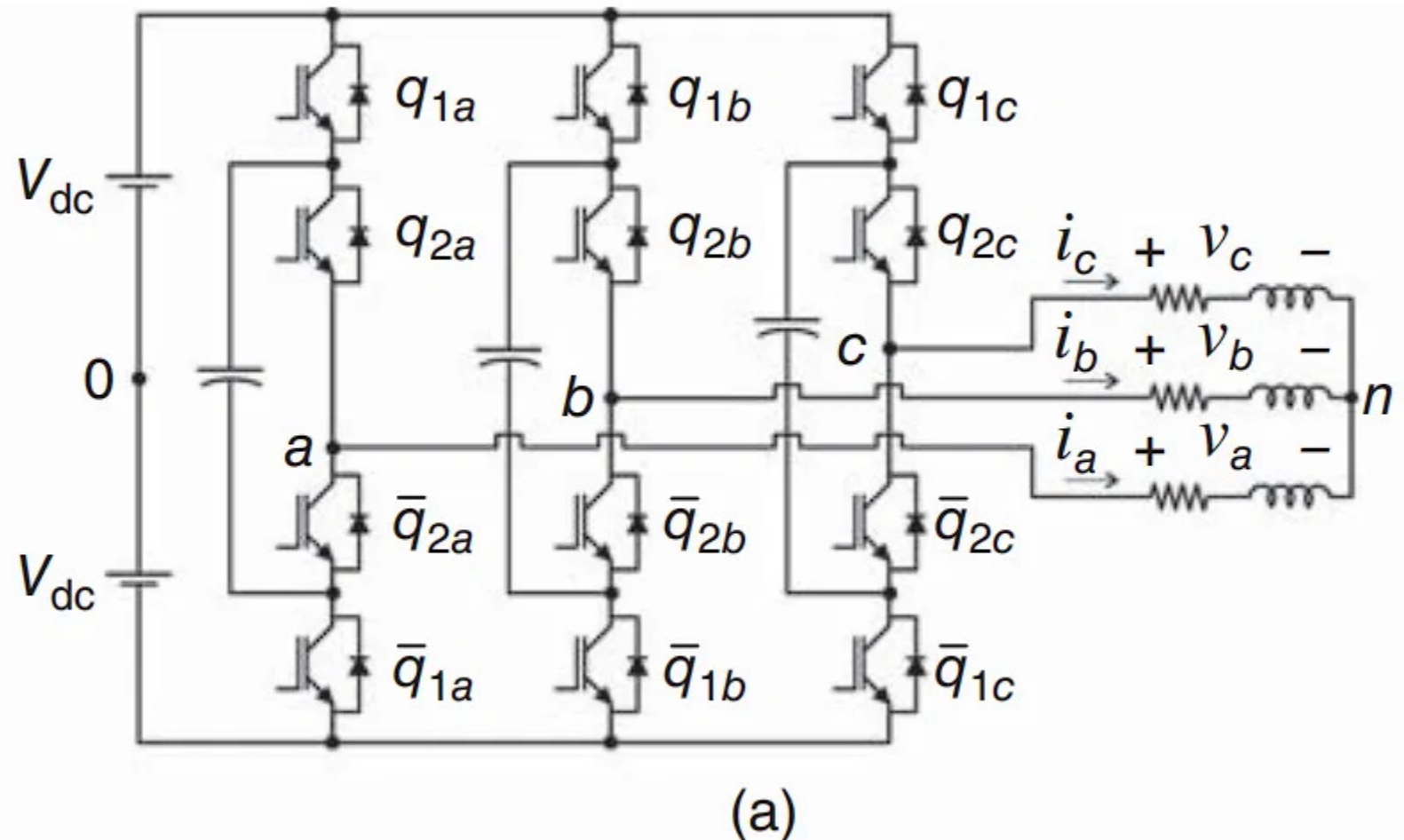
Output voltage levels: The topology synthesizes three distinct voltage levels per phase relative to the DC midpoint (0): $+V_{dc}$, 0, and $-V_{dc}$.

Redundant zero states enable natural balancing:

State 1: $q_1=1, q_2=0 \rightarrow$
Output = 0V (Charges C_{fc})

State 2: $q_1=0, q_2=1 \rightarrow$
Output = 0V (Discharges C_{fc})

PD-PWM modulation:
Phase Disposition PWM actively selects between the redundant zero states based on the phase current direction to maintain V_{fc} at exactly $V_{dc}/2$.



Source: Advanced Power Electronics Converters (E. Cipriano)

Component Selection

Switch: Wolfspeed

CAB450M12XM3 — 1200V rating provides a 50% margin over the 400V/switch stress. 450A rating offers a 3.3× margin over the 136A peak current.

SiC chosen over Si IGBT — Delivers 10× lower switching energy, eliminates tail current, and enables efficient 20 kHz operation.

Capacitor sizing — $C = I_{pk} / (2 \times f_{sw_min} \times \Delta V_{fc}) = 136A / (2 \times 5kHz \times 20V) = 680\mu F$.

Film capacitor (polypropylene) — Selected for $ESR < 5m\Omega$, >100A RMS ripple capability, self-healing properties, and automotive-grade reliability.

Thermal modeling — $R_{th_jc} = 0.094\text{ }^{\circ}C/W$, modeled using a 4-layer Foster network for accurate transient thermal analysis.



Vehicle Architecture Mapping

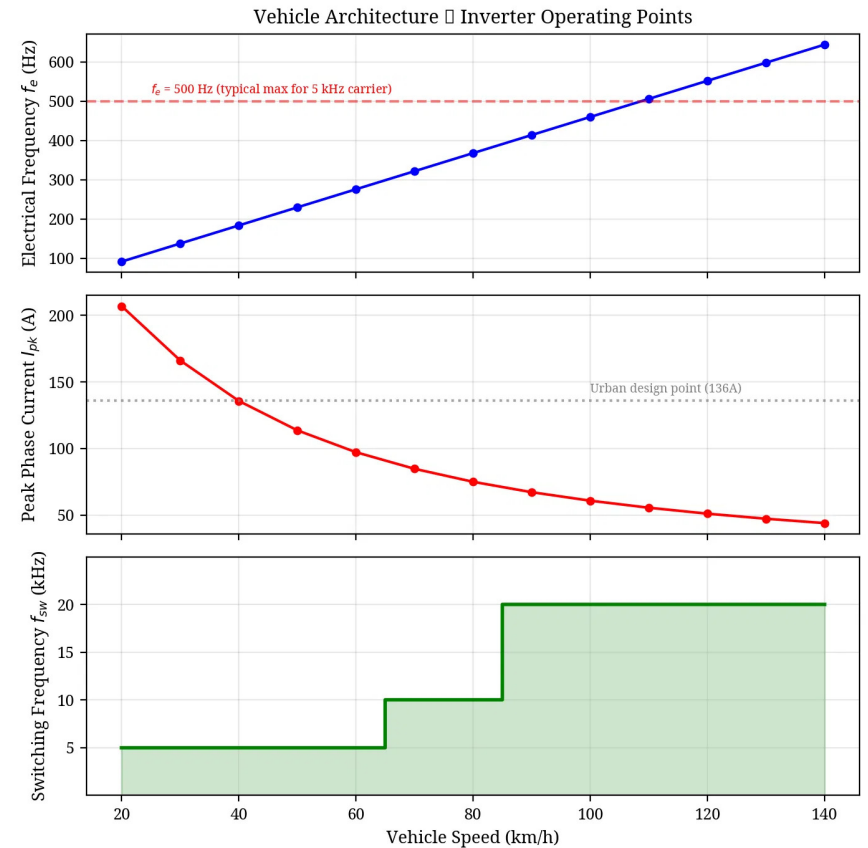
Drivetrain parameters dictate the operating points: **G = 8.19** gear ratio, **r = 0.315m** tyre radius, **p = 4** pole pairs.

Electrical frequency (f_e) is directly proportional to vehicle speed (v):

$$f_e = (v/r) \times (G/2\pi) \times p$$

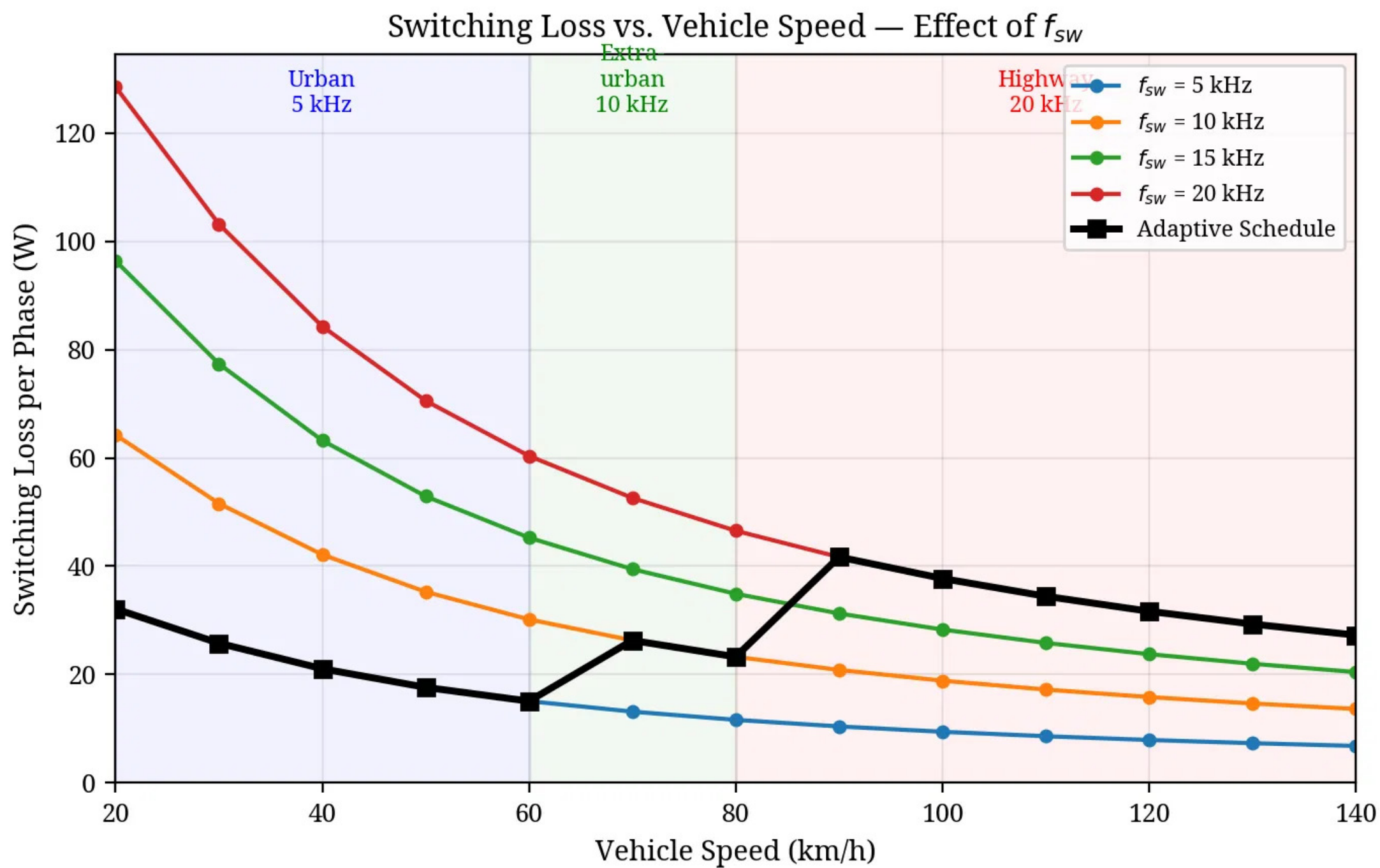
Using an RL load model ($R = 1.3\Omega$, $L = 2\text{mH}$), **current decreases with speed** as load impedance increases.

- **Urban (40 km/h):** $f_e = 184\text{ Hz}$, $I_{pk} = 136\text{ A}$ — highest thermal stress.
- **Highway (120 km/h):** $f_e = 552\text{ Hz}$, $I_{pk} = 51\text{ A}$ — lowest loss but highest frequency demand.



Switching Loss Analysis

Graovac-Purschel analytical method used for loss calculation:
 $P_{sw} = N_c \times (f_{sw}/\pi) \times E_{sw_ref} \times (I_{pk}/I_{ref}) \times (V_{dc_sw}/V_{ref})$
Urban driving at 20 kHz:
 $P_{sw} = 84.3$ W/phase vs $P_{cond} = 51.5$ W/phase. Switching losses heavily dominate (63% of total) due to high phase currents (136A).
Highway driving at 20 kHz: $P_{sw} = 31.7$ W/phase vs $P_{cond} = 5.6$ W/phase. Both losses are relatively low due to reduced current (51A).
Key insight: Operating at 20 kHz is unnecessary at 40 km/h where the fundamental electrical frequency (f_e) is only 184 Hz. A minimum carrier ratio ($m_f = f_{sw}/f_e$) of 27 at 5 kHz / 184 Hz is sufficient to maintain excellent THD while drastically cutting losses.



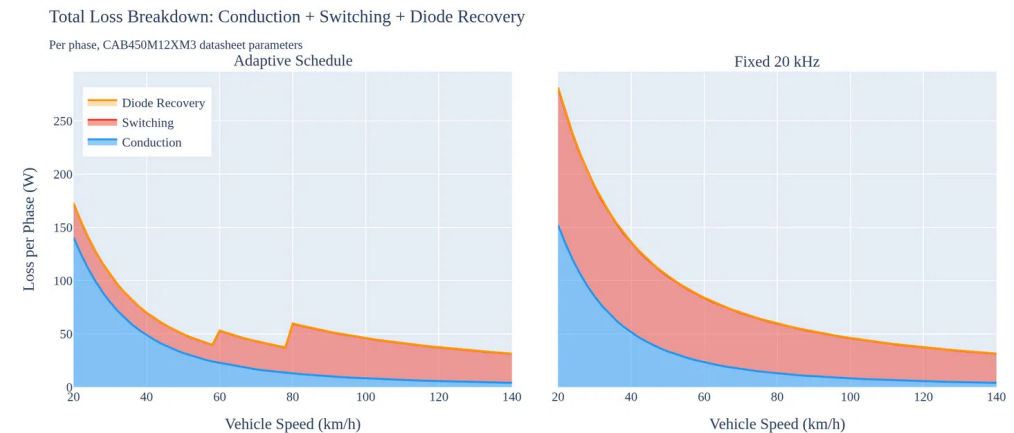
Simulated Adaptive Switching Frequency

An adaptive switching frequency schedule was **fully implemented and validated within the dynamic time-domain simulation**.

The methodology relies on maintaining a sufficient **frequency modulation ratio ($m_f = f_{sw} / f_e$)** to ensure waveform quality while minimizing losses:

Urban Driving (Low f_e): At 40 km/h, f_e is only 184 Hz. The simulated controller dynamically drops f_{sw} to 5 kHz while keeping $m_f \approx 27$. This yields a massive 75% reduction in switching losses exactly when the phase current is highest (136 A).

Highway Driving (High f_e): At 120 km/h, f_e reaches 552 Hz. The controller automatically increases f_{sw} to 20 kHz to maintain $m_f \approx 36$ and prevent distortion. Fortunately, the phase current is much lower (51 A), keeping absolute losses manageable.



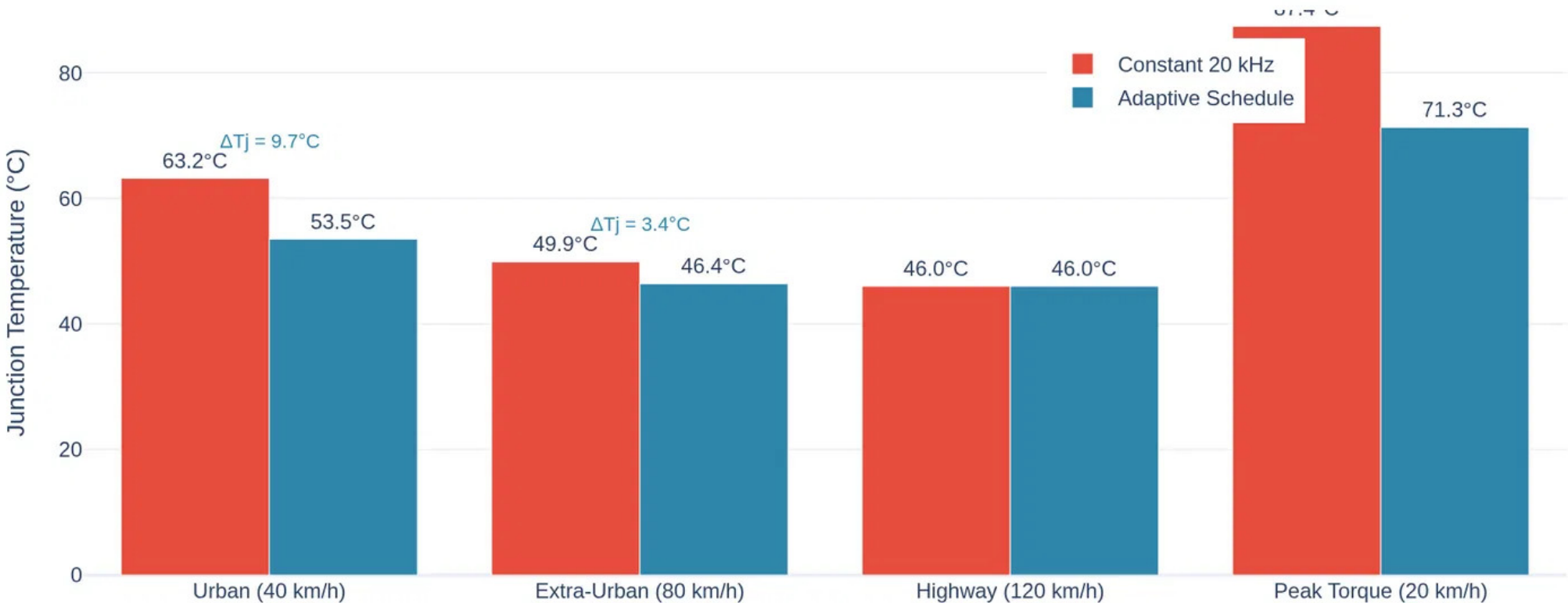
Thermal Impact of Adaptive Schedule

Urban Tj reduction: Junction temperature drops from 63.2°C (fixed 20 kHz) to 53.5°C (adaptive 5 kHz) — a **ΔTj of 9.7°C**.

Peak torque Tj reduction: At 20 km/h, the temperature drops from 87.4°C to 71.3°C — a massive **ΔTj of 16.2°C**. **Thermal cycling amplitude** over a mixed drive cycle is reduced by 53% (from 16.4°C to 7.7°C). Per the Coffin-Manson fatigue model, this reduced thermal cycling directly and significantly **extends solder fatigue lifetime**.

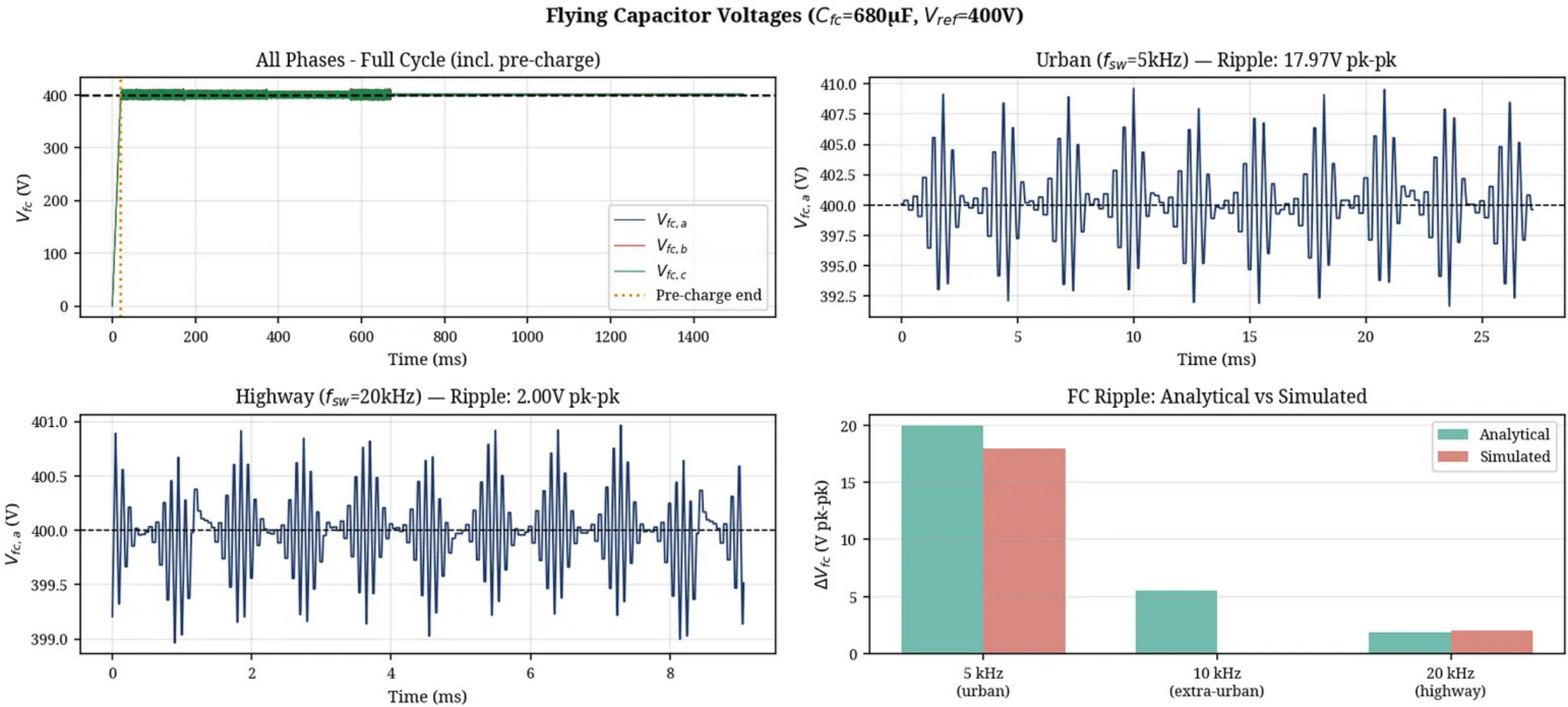
All operating points remain well below the module's Tj_max of 175°C, providing **>100°C of thermal margin**.

Junction Temperature at Key Operating Points



Simulation Results

Time-domain simulation confirms proper **3-level output voltage** synthesis at both urban and highway speeds. Phase currents are clean sinusoids with **THD < 2%** (1.92% at 5 kHz urban, 1.13% at 20 kHz highway). FC voltage is successfully maintained at 400V with **$\pm 0.04\text{V}$ mean error** using PD-PWM active state selection. FC ripple: **18V simulated** vs 20V analytical budget — excellent agreement with design calculations. **Pre-charge transient:** 20ms linear ramp from 0V to 400V successfully executed before active switching commences.



Conclusion

800V architecture is justified by ultra-fast charging speed, reduced cable weight, and overall efficiency gains.

FC topology was selected over NPC for its superior thermal symmetry and natural balancing capabilities under dynamic EV loads.

Adaptive fsw schedule: Achieved a 75% switching loss reduction in urban driving, saving 190W (3-phase) compared to a fixed 20 kHz baseline.

Thermal benefit: Delivered a 9.7°C Tj reduction and 53% less thermal cycling, directly extending the power module's fatigue lifetime.

All metrics validated: Time-domain simulation confirmed THD < 2%, FC balance < 0.04V error, and Tj < 54°C with >120°C thermal margin.

Future work: Experimental validation of the adaptive schedule and thermal models on a hardware prototype.